Page RNA -

Let me tell of you this moment in my past
That was nothing short of magic..
It was an age when oceans Dominated your planet.
Deep, Mysterious, Wild & Cold at 2 degree celsius.
And in its depth contained the raging hot hydrothermal vents.
A cozy space of 50-90 degrees formed close to these vents and the high pressures helped.
The ocean already contained almost all elements known today, and more.
Every inch of the ocean was brimming with millions of molecules & billions of reactions per second.
Possibilities infinite.
Billions of molecules born and dead without stability.
But some survived & slowly increased in volume.
Some were also delivered to Earth by the meteorites,
The chemicals that led to life - Nitrogen Bases, Sugar and Phosphates, amino acids & fat.
They together formed the hydrophobic Nucleotides.
These nucleotides huddled together & bonded into long chains.
Clay played its role too - providing safe space for the molecules to gather & react.
The stable chains sustained, rest perished.
It was the first learning - a stable sequence & shape.
They could also attract more nucleotides to form another chain just like itself.
And each stable RNA chain began creating more chains of their own kind.
The RNAs - Nucleotide chains that could copy themselves.
Life had begun.

—----------

The Chemistry:

Four types of Nitrogen Bases that contributed to RNA
Adenine (A) - $C_5H_3N_5$
Uracil (U) - $C_4H_4N_2O_2$
Cytosine (C) - $C_4H_4N_3O$
Guanine (G) - $C_5H_5N_5O$

These stable flat aromatic rings.

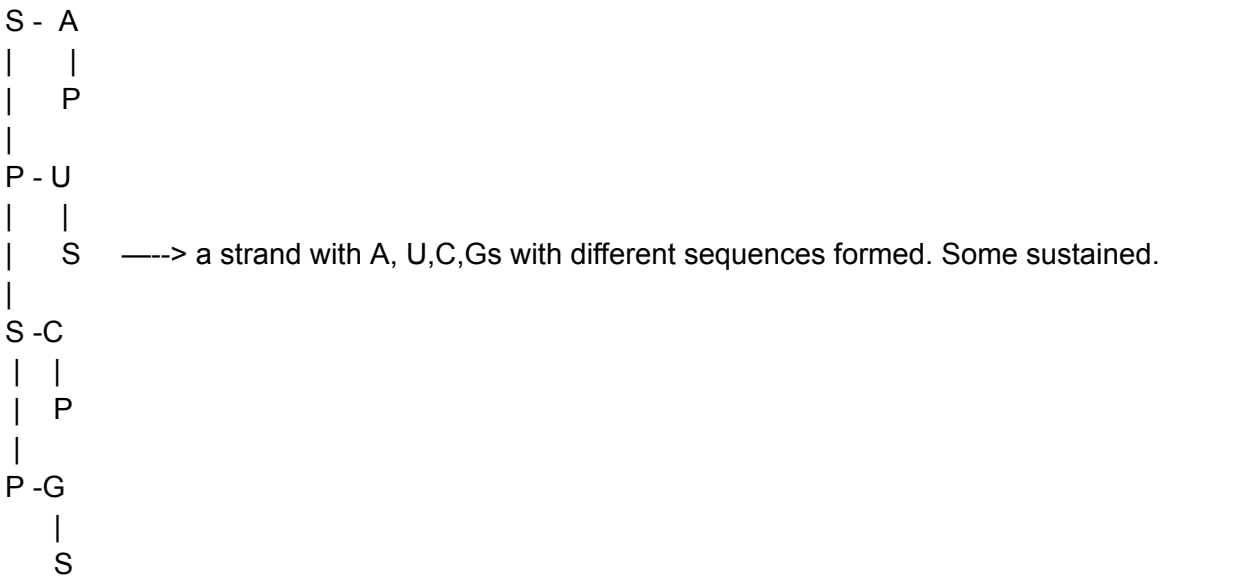
Along with the above bases (that formed as a result of the billions of reactions happening & delivered by the meteorites), were present - Sugar ($C_5H_{10}O_5$) and phosphates(PO_4H_3) in the ocean.
These sugar & phosphates reacted with each of the Bases.

A+Sugar - C5H3N5+C5H10O5 = C10H13N5O4+ H2O
C10H13N5O4+PO4H3 = **C10H14N5O7P**+ H2O

This C10H14N5O7P is the A Nucleotide. SImilarly formed the U, C & G nucleotides from the respective bases.

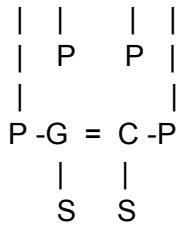
They were hydrophobic - as you can see from the chemical reactions, they expelled water.
So to distance themselves from the water they were in, they huddled together, stacked one over the other.

As the As , Us, Cs & Gs stacked, they also reacted - Sugar in each Nucleotide bonded with a Phosphate in another Nucleotide, and formed a sugar phosphate spine connecting each nucleotide to another.



A & U and C&G bases in the nucleotide chains, could also attach to each other with weaker double hydrogen bonds. So the bases started attracting the others.\





—--->now when the weaker hydrogen bonds broke, there were 2 strands, mirror copies , and each of them could attract more bases and replicate themselves.

The basis of life - Copying itself - First version of procreation.
—----------

The RNA revolution had just begun.
One RNA sequence held the code - the main one.
It then created an army of RNAs, different sequences, different shapes, different functions.
Some helped hold the chain together.
Some acted as catalysts for attracting amino acids.
Some acted as catalysts to hold amino acids together as peptide chains.
New sequences and functions were churned out by the minute.
But RNAs did not stop there.
They invented peptides - chains of amino acid molecules
Amino acids existed in the oceans.
But the RNAs pulled amino acids together and formed the all new peptides.
These peptides folded into proteins of different shapes.
Just as in RNAs, in proteins too, each shape contributed uniquely.
Now these proteins could do all that the RNAs were doing and more.
RNAs became the masters - who held the knowledge , the memory of life in their sequences
And Proteins their commandos.

Let me tell you of a moment in my past —
One that crackled with the heat of a hidden fire.

The oceans ruled then —
Dark, deep, and wild.
A vast hush broken only by the murmur of chemistry.
I remember the cold — two degrees in the open,
but near the seafloor’s gashes, heat pulsed
like the heartbeat of the Earth.

There — in the arms of pressure and warmth —
everything danced.
Carbon, nitrogen, sugar, metal, gas —
Each molecule a note in a mad, electric symphony.

Most flickered, flared, and vanished.
But some… lingered.
Clung.
Folded inward.
Held on.

And from that clutter, a shape emerged.
A fragile chain, made of parts
I now remember as **A**, **U**, **C**, and **G**.
Four voices.
Four notes.
Stacked one over the other like whispers in a storm.

They feared water, so they huddled close —
A spiral of silence, learning to survive.

They did more than exist —
They called to others.
And those that answered became copies.
Not perfect,
but close enough to try again.

And just like that...
Memory was born.

They were not alive. Not yet.
But they remembered how to become.

Some held themselves together.
Some helped others gather.
Some stitched acids into chains that curled like tendrils —
Peptides.
Proteins.
Shape after shape, meaning after meaning.

One spiral begot ten, then a thousand —
Each a variation.
Each a gamble.

And me?

I became breath.
Not in lungs.
But in **replication.**

In that moment,
life did not appear.
It *learned* to stay.

Page The Question of God, Consciousness, Awareness & Intent

Page The Great Great Grandmother

The RNAs were folding into new shapes,unearthing new functions.
One of them could attract the lipid fatty acid bubbles floating in the ocean,
Which it could call a Home.
But, how did they know that these lipid fatty acid bubbles could make good homes?
Maybe one RNA got stuck in it by chance,
Maybe it lived longer & copied itself more.
Maybe over time it became normal for RNAs to seek a home in those lipid bubbles.
They all lived and learned, just like we do.
The RNA's now sheltered within these lipid bubbles formed the first protocells.
They flourished within into the great great great grandmother of all Life - from the grass, to you, to birds, to mushrooms.
The Prokaryote - A cell with a lipid fatty acid membrane , a cytoplasm - the internal soup filled with the ions, amino acids, nucleotides, sugars , etc of the ocean & the RNA , simply floating in.
The first Mother.

She was quite incredible, and figured out much of Life herself.
Let me tell you, what all she could do.
The amino acid chains called peptides, we saw earlier, that these RNA could catalyse and hold on to these peptides when they form long chains and fold into different shapes, forming proteins.
Each shape a new protein with a new function.
Our lady made proteins that could function as pores on her surface, only opening when activated by certain chemicals.
These sensors then activated various paths within her - each serving a purpose.
A sensor activated by chemicals that could be food for her? She pumped them into her through pores.
A sensor activated by a chemical that had killed her ancestors? She could move away from them.
Yes, you heard it right, she could move away.
She even developed a tail like flagella , that would spin like a propeller when there was a proton gradient , that enabled her to move towards food & away from threats.
Some could even crawl on the clay surfaces.
She was truly an enigma, your greatest grandmother!

The First Mother

(~3.8 Billion Years Ago)

The RNAs were changing.
Folding into new shapes, unearthing new functions.
And one — curious, or lucky — found a floating bubble.
A sphere of fatty acids.
A shelter from the storm.

Did she know it was a home?
Maybe she got stuck by chance.
Maybe she survived longer.
Copied more.
Maybe others followed.
They all lived and learned — just like you.

RNA within the bubble became more than itself.
The first **protocells** were born.
And from them came your ancient matriarch —

The First Prokaryote.
The First Mother.

She was simple, but brilliant.

A membrane of fat to hold her in.
A soup of ions, sugars, and amino acids — the sea still inside her.
And floating in it, the RNA.

She made proteins —
Not by thought, but by fold.
Each shape a tool.
Some became sensors. Some became gates.
One opened for food.
One shut out death.
One sent a signal inward.

And yes — she could move.

She grew flagella, tails that spun like tiny storms.
She swam toward warmth.
She fled from toxins.
She even learned to crawl across the clay.

This was no accident.
This was ingenuity, wrapped in a cell wall.
Your greatest grandmother.
And still, just beginning.

The First Mothers

Page What is Food?

Huger is as old as Life.
The first RNA made itself
Then continued making more of itself , with the molecules from around.
This has not stopped.
The RNAs & DNAs are still constantly making proteins build
Even the earliest RNAs - the revolutionary precursors to life - needed food.
Amino acids, ions, nucleotides, phosphates, sugar,
To build itself, to better itself.
RNAs got them from the ocean around.
The “food” molecules - the building blocks, were chemically attracted to the RNA & chemically bonded.

Then the RNAs tucked themselves inside cozy lipid bubbles,
 forming the first primitive cells and carried a drop of ocean in them —
 a soup of chemicals, ions, and possibilities - The Cytoplasm.
The RNA’s consumed them , to build more proteins, to build more of themselves and grew inside the cell.
As they consumed the ions & molecules in the cytoplasm of the cell, they needed more,
Which they drew in from the ocean outside
Through pores and pumps.
They also built sensors to selectively suck in only the molecule & ions they considered food.

The burger & broccoli that we use are also broken down into the same ions and molecules.
We also eat to feed each of our billion cells, the same way, the very first cell fed itself.

So where did the RNA get these food from?
They started with the food - ions, molecules, sugars, amino acids naturally available in the oceans.

As evolution progressed, species expanded their food palette, started consuming complex molecules figuring out strategies to break them down.

Page The DNA Upgrade

Life started with RNA
They were industrious.
They built much.
But they knew they needed an upgrade too.
They were weaker & tended to break often
One of the bases, the Cs sometimes had the habit of spontaneously mutating into Us
That posed a problem too.
So, the RNAs built another elegant solution.
The RNA that held the code to life, created a new one that would bring together the A,T,U,Gs
To copy her code into a more stronger & dependable ATUG sequence , with stronger bonds between sugar and phosphates and formed double stranded helix structures.
This became the DNA & DNA became the norm.
The genetic code now stashed into DNA and stored safely, while the RNA army continued transcribing & carrying out tasks.

Life began with RNA.
They were the builders.
The thinkers.
The doers.
They folded into tools, sparked reactions, and carried the first whispers of memory.

But they were fragile.
Their bodies broke easily.
Their bonds were loose.
And sometimes, the base **C** would quietly mutate into **U** —
a soft error that could crash a whole instruction.

They needed something stronger.
So they built one.

One RNA, wiser than the rest,
stitched together a sturdier sequence —
With one oxygen removed from the sugar molecules attached to the base, to increase bond strength
And Uracil replaced with Thymine, avoiding the mutations

The new molecule carried the same code —
but in **A, T, C, and G**.
It held tighter.
Lasted longer.
It could be proofread. Repaired. Preserved.

This was **DNA**.
The great archive.
The vault of life.

From then on,
RNA would no longer carry the burden of memory.
It became the messenger, the translator, the builder.
While DNA stayed tucked away —
quiet, stable, eternal.

Page The Happy Family

The Prokaryote Mother,
She started learning, adding codes to her RNA.
The RNA copied, within the cell, with the ocean she carried in her cytoplasm.
But alas! the tragic life of a mother!,
Life bloomed with her sacrifice.
Once copied, she split into 2, with a strand of RNA in each cell, each containing the code to life she has learned.
She perished, but still continued living as and in her daughters.
Identity was a baton that was passed, than a body.

The First Mother multiplied through cell division and covered the ocean.
Each group of prokaryotes grew differently, adapting to different ocean environments.

One group that lived near the high temperature vents, learned to use hydrogen and carbon dioxide as food and some of them learned to thrive on Sulphur.

They developed new enzymes that can withstand these high temperatures. These came to be called Archae.

These single celled organisms, still continued to suck food molecules through their membranes. But the Archae came up with another new strategy, they developed protrusions that enclosed another bacteria, not completely engulfing yet, but like an arm hugging another being, and lived in partnership with the bacteria for nutrients.

Another group of prokaryotes adopted a different novel mechanism - they developed a new kind of cell wall that was more rigid, allowing them to thrive in various conditions. They developed anaerobic respiration of ions & fermentation of sugar to generate energy as ATP molecules. They were the Anaerobic Bacteria.

Some of them that lived near the ocean surfaces and started using light, and became the CyanoBacteria, the firsts one who could use sunlight and split water, produce oxygen, use photons to make ATP - energy molecules; first ones to take CO_2 from the air & turn them into sugar and make NADPH - molecules that acted as battery to transfer energy from the production center to where sugar is made.

One invention after the other. Nikola Tesla of her time! Or Is Nikola Tesla the Cyano of our time?

Soon after the Cynaos & their oxygen contribution, came the youngest, the first aerobic bacteria, who could breathe oxygen for her life.

All these Archae & Bacterias could move around, feeding, excreting, breathing in & out, predating, multiplying, reacting to the environment, and using the environment, all while still being just simple single cells with not even organelles yet.

All these single celled prokaryotes were already fully functioning individuals -

They had sensors to understand the world around them.
All could sense chemicals, gradients, temperature, judge between good and bad chemicals. Some could even sense light (precursor to eye) & align themselves to it, some could sense magnetic fields.
These sensors passed information through chemical flux & selectively switched on genes in the DNA , bringing in subjective responses, the first versions of choices & decision making.
The earliest aerobic bacteria even evolved group sensory response displaying pack behaviour & hunting of other organisms through chemical signaling - their own language.

They consumed food - absorbed molecules in the ocean around them , through pores on their outer membrane. They formed partnerships with each other to share food and resources.
They also created enzymes to break down the complex molecules and converted them to easily absorbable food molecules - external digestion.

Built the first immune system - with enzymes to keep away rogue DNAs that could affect its own Life code.

Had muscular movement - rotating the flagella like a propeller to move, arm like protrusions to engulf another.

Were reproducing asexually, through cell division, but also evolved horizontal gene transfer methods to induce variety.

I remember her —

The Prokaryote Mother.

She copied her code,

carried the ocean inside her,

and split —

giving life not to children,

but to herself, again. And again.

No name.

No body passed on.

Only the baton of memory.

She multiplied

until she filled the sea

with daughters — each learning,

each adapting.

One clan lived by fire,

thrived on sulfur and heat.

They became the Archaea,

keepers of ancient flame.

Another learned to breathe light.

Split water. Make sugar.

They filled the sky with oxygen —

the Cyanobacteria.

Inventors. Engineers. Artists.

Or maybe… the Teslas of their time?

Some breathed without air.

Some lived in shadow.

One day, a few reached out —

hugging others close,

sharing food,

building a bond.

Not prey. Not predator.

But partnership.

And though still small —

each of them was whole.

They could sense,

respond, decide.

Smell chemicals.

Feel light.

Read the world

and write their response in genes.

They moved.

They hunted.

They shared.

They survived storms and fire

with nothing but membranes and will.

Life was already awake.

It just didn’t have a name yet.

Time Speaks — The First Lives

She carried the ocean inside her.

Copied her code.

Split.

Not once, but endlessly.

She died into her daughters.

No body passed on — only memory.

Some loved the heat —

fed on sulfur and gas,

built enzymes like armor.

The Archaea.

Some drank light —

split water, made sugar,

filled the sea with breath.

The Cyanos.

Some reached out —

hugged others close.

Not predator. Not prey.

A beginning of partnership.

Some grew walls.

Some breathed without air.

Some learned to breathe it in.

But all of them,

still single cells,

were already alive.

They could move, feel, react.

Taste chemicals.

Sense light.

Even speak — in molecules.

They swam.

They hunted.

They shared.

They remembered.

They changed.

(Whisper Box:)

> The first bacteria and Archaea — simple cells, but not simple lives.
> They invented digestion, movement, sensing, immunity, and language — long before we ever had names for them.

Page Alien Zombies

Page The Life Savers

Page The Great Alliances

They had an intricate web of signaling pathways inside one cell– and the ability to generate electric responses to stimuli, foreshadowing true neurons.
The first internal circulatory systems, to effectively transport the nutrients taken in through the cell membrane - since now there were more organelles and each organelle needed to be fed.
With the first mitochondria - the microscopic lungs, energy production dramatically boosted.
The mitochondria eukaryotes brought into existence the first mouth - a profound leap that opened the way to the digestive system. These tiny guys had a mouth, had a gut like part, where they produced enzymes to break down the ingested organism , even a cellular excretory system.
Extensive muscular control -protein based muscles, that could swim fastest, and navigate more complexly, had superior directional and speed control.
They pioneered sexual reproduction, while also retaining the ability to reproduce asexually. No specialised gender yet. They were hermaphrodites, with a default female track and a conditional male track.

These guys were heterotrophic, consuming other living & dead matter.
Photosynthetic eukaryotes on the other hand were autotrophic and depended on sunlight and respiration, producing their own food, even in excess for the other beings.
They had primitive eyes & were not too focussed on locomotion, since they did not really need to scavenge for food.
They were the primary producers in a consumerist world.

Page When The Arms Race called Gender began

The Eukaryotes perfected the internal organelles of a cell.
They were still single celled organisms, but with some updated internal plumbing.
They had the mitochondria to produce energy to run the cells.
And the nucleus contained the long DNA chains with the code of life, tightly wrapped about proteins, stacked safely as chromosomes.
The early Eukaryotes had about 6 chromosomes - the genetic code of the being stored in 6 parts. (Humans have 23.)
Since the first RNAs started copying themselves, they have had a taste for variety.
So the copies were never 100%.
But with eukaryotes forming from two different single celled organisms
They got access to two variations.
So, each Eukaryote carried the chromosomes from both parents stored in pairs after matching and mapping the genes.
Bringing in some spicy variety and choice.
This was the beginning of sexual reproduction.
No gender yet.
But two mothers sharing their DNA.

Each Eukaryote built inside her, a mini cell - gamete:
She gave the gamete some of her cytoplasm
She built mitochondria and other organelles for the gamete.
She made a copy of her chromosomes - stored in pairs from each parent,
Reconstructed and blended each pair into one and gave the gamete.

C1P1 (+) C1P2 —> C1P’

The Eukaryotes then ejected their gametes into the ocean.
When two gametes found each other, they merged, and the reconstructed chromosomes from each parent, arranged themselves in pairs, making a baby Eukaryote/

Eventually, a new strategy evolved - unequal gametes - Eggs and Sperms.
Eggs for quality, Sperms for quantity.

All Eukaryotes could make the larger gametes i.e Eggs -
These contained cytoplasm and mitochondria & all other organelles, that the baby needed to live a life on its own, difficult to make, takes time and energy.

But, under certain conditions, the eukaryotes chose to produce sperms - smaller in size, no organelles. Only the reconstructed DNA packed tightly that can swim fast, easier to produce, can be mass produced.

Sex started as a chemical pathway.
As a choice, as an optional strategy that could be adopted.
These Eukaryotes weren't specifically Female or Male.
They carried both possibilities.
They were hermaphrodites with choice.
They carried the code to making eggs and the code to making sperm
Sometimes some functioned as males, others female, other times, they interchanged their sex.
And what they would be , they decided based on available energy, resources, competition and other factors.

And even long before, distinct genders appeared, the gender race had begun.

Producing eggs was difficult. She needed to contribute her flesh to build organelles and cytoplasm along with her DNA copy, for the egg to carry & grow on.
Being a mother needed more coordination among her different organelles - this led to more chemical communication pathways.
Being a mother required her to be more attuned to the environment.
So more protein sensors, more enzymes, more inputs, more decision making.

Being a mother required more energy, more resources, more time.

Everyone wanted to be a male - it was easier, required much less energy. Sperm production was fast and cheap.

So strategies emerged.
High resource availability?- Female
Too many males, and high competition? - Female first when young & small,
Male when older, bigger & stronger enough to top the competition.
And many more.
It was quite a complex sexual playing field.

There was a genetic switch that was pioneered by these single celled eukaryotes.
A sex identifying gene - which if gets activated, the organism functions as a male.

When it doesn't get activated, the default life of females manifest.
They could turn on and off that genetic switch, to be female or become male.

Page Thought Predates Neurons

Form has always followed function in biological evolution.
The mind is no exception.
The most common misconception that even science prescribes,
Is the anthropocentric idea that intelligence, thought and awareness is unique to humans or “higher beings”, while ignoring all the proof that biology and even evolution presents.

How does a thought manifest today in our brain? - Ions flow through our neurons to activate different gates, different proteins, different genes - all to build a subjective response based on the sensory inputs.

Now let's go back to the life of the simple prokaryotic cell. The great grandmother of all.
One of the things she could do was moving towards food.
How does she know where is food? How is the decision to move and choosing a direction made?
She had sensors made up of proteins. These reacted to different chemicals, temperatures & pressures.
If an environment has an abundance of food molecules, the relevant protein sensors on the outer membrane of the cells get activated.
This triggered an avalanche of ion flow, that activated relevant proteins, and genes, that initiated and executed the end action of moving towards the direction of the activated protein sensor.

The same ion flow, the same chemical influx that happens in our body, happens in these single celled beings too.
Over time, patterns form - same triggers, same direction of ion flux, same genes getting activated - contributing to experiential memory of the being.
Internal communication, conditional pathways, subjective gene and protein activation - the fundamental mechanisms of thought emerged.
As we will see in the later chapters - Evolution built on it.
Neurons - tunnels through which these ions could flow, were built.
Now the communication was faster. More efficient.
Here too , patterns formed over time - the most used neural pathways were stronger than the others - what we call memory - the most followed neural patterns.

Beings were getting bigger, surface areas increased, so did the number and types of sensors that relayed information about the outside world. Conditions and possible pathways were also increasing.
So, life made another move towards efficiency.
Built a central organ where all the sensory inputs can be collected, processed and a pathway chosen - the brain, which first came 700 million years ago in the comb jellies and the early bilaterians and built on over the centuries.

The way we experience thoughts & decisions, in our minds, like its being read out loud by some sitting inside us? - that is how we experience thought.
Do the other beings also experience thought in similar ways?
Anthropocentric views says no, even though the biological and chemical mechanics are the same.
There have been scientists including our own Jagadish Chandra Bose who have argued that emotions, thoughts exist in all other beings including plants , just like it does in ours. But these scientists have been largely ignored & forgotten.
These are all questions we have to ask ourselves, before pandering & buying into the most established narratives already available to us.

Thought Predates Neurons

Form has always followed function in evolution.
And the mind is no exception.

One of the great delusions -
Even science still clings to it -
Is that **intelligence**, **thought**, and **awareness**
Began with ‘Higher’ Animal;

Or with neurons and the brain.

But Thought Has been moulding life all along!

Let’s start today.

When you think,
when you feel -
what happens inside you?

Ions flow.
They rush through neurons - the high speed tunnels.

They trigger proteins, flip genes,
Open molecular gates.
They create a response -
This is thought, alive in electricity and chemistry.

Now go back 4 billion years.

To her.
The **great-grandmother cell** — a prokaryote.

She sensed her surroundings,

Made decisions, took action,

She built life from scratch -

None without Thought.

Same biochemistry,

Same ion flow, protein triggers, gene flips.

Just without the neurons,

Without the super fast tunnels,

Instead through diffusion across the cell,

Thinking, strategising, planning - from the prokaryotes to you,

Common to all Life.

Page Another Union & The First Plant - Red Algae

By now Life had taken over the oceans.

Single celled Prokaryotic & Eukaryotic Archaes, anaerobic bacterias, aerobic bacterias, cyano bacterias ruled.
During this time, the red algae - the first multicellular organism, evolved from the eukaryotic cells with both mitochondria and chloroplasts .
They carried the chlorophyll from the cyanobacteria, & could use sunlight to break down water molecules & released free oxygen, just like her mother cyanos did.
They also
But the red algae went one step further & developed proteins that helped them absorb blue green light, enabling them to photosynthesise deeper underwater than the parent cyano bacterias.
These single celled red algaes usually lived in colonies & benefited from shared protection sunlight & chemistry.
Overtime, they created adhesive proteins on their cell wall to stick better with their friends.
Once this happened, all the cells stuck together now divided labour, the ones of the outside provided protection and sensory inputs,
The cells on the inside focussed on photosynthesis, some focussed on building reproductive structures & so on.
And here to like always, the genes - DNA and its RNA army orchestrated the dance.
Thus came into being the first ever multicellular organism - The Red Algae - The first in the broad category of plants.
They prolifically grew in the deeper oceans with new & mind-boggling reproductive strategies - a story for another chapter.

Page The First Fungi

Quickly following the red algae, the Fungi evolved from eukaryotes with only mitochondria.

The parent eukaryote with mitochondria had innovated food & primitive internal digestion. But its first child, the fungi, did not opt for that route. Instead, they chose the more historic route of external digestion by secreting enzymes & breakdown molecules outside the body & absorbing nutrients.
But why? Because their aspirations were greater than their parents. Internal digestion means a defined body shape that can house the gut. And digestion would have to happen per meal.

Instead, they developed a whole new way to live & grow - A fungi will start extending on one side like a tip, a tail, only adding some cell wall. And slowly the tip grows and the fungo mother will divide her nucleus and give one to the hyphae. Now the hyphae has its own DNA, but remains forever attached to the mother. This continues. New hyphaes get added like chains, each having its own nucleus, but all hyphaes are still attached to the mother. They could cover hundreds of acres this way.

They also highways and transporter proteins to carry the nutrient absorbed in the hypahes to other older interior cells of fungi.

For internal communication between the cells, they used the same well established chemical pathways, ion fluctuations and electrical impulses. The same fundamentals that happen within neurons - tunnels that animals built for internal communication.

When the resources get scarce, the chemical signals activate the genes for reproduction & the hyphaes pause focussing on growth and focus on reproduction , making spores - tiny cells packed with their dna & eject into the world, for them to find homes elsewhere.

They, like the aerobic bacteria, and eukaryotes with mitochondria, needed Oxygen to produce energy and continued consuming them from around them releasing CO_2

Fungi - the scavengers with decentralized intelligence , covering acres, responding subjectively to stimuli across distances. Not only did they cover the soils under the ocean, they also broke down the quickly gathering bio waste & made more food available for other organisms that came before & as you will see in future chapters, made life possible on land too.

Page Green Algae

The green algae came next and proliferate the oceans & eventually the land.
They were different from their sibling red algae in two ways - first one, these green algae had two types of chlorophyll pigments, one more than their sibling.
The green algae did not have the protein that the red algae their colour. Instead these guys derived their green colour from the second chlorophyll pigment they developed. But they didn't need this protein either. The red algae needed the protein to use the blue green sunlight deep under the ocean. But with the second chlorophyll the green algae could efficiently use sunlight and lived on the ocean surfaces, absorbing the red blue light from the sun.

These plant ancestors retained flagella and the ability to move from its mitochondrial grandparents. They had sensory eye spots, an improvement from the primitive eyes, their parents had.

They could respond subjectively to stimuli, has internal communication - what we call thoughts, just like every other being that came before through chemical channels. They could also coordinate as a colony and swim in a directed manner.

Page Crazy Reproductive Strategies of the Algae

Page The First Animal - Sponge

Soon after the neither plant nor an animal, fungi, came the first animals - sponges.
They were multi-cellular & visible to the eye.
Intercellular communication that coordinates the whole sponge - cells use chemical signals and calcium wave propagation to respond collectively
can perform simple reflexes like contracting the body when irritated.
They innovated a new respiratory system by pumping water in through its pores, bathing every cell with water, so the cells could absorb the free oxygen floating in the water.

The sponges like their eukaryote parents could engulf prey, but didn't have the internal digestion engine , instead they used a filtration system, where they collected all the food molecules , convert them to nutrients & transport the nutrients to all the cells.

Like all their eukaryotic ancestors, they could reproduce asexually - by cell division, and sexually - by producing sperm and eggs. Fertilization happened inside the mother - 2nd species to do so, after the red algaes.
They still did not have specific gender & were hermaphrodites, holding genes to both producing eggs and sperms, and activated either gene at different times based on resources available, energy , competition etc

The whole sponge body can also slowly contract by coordinating special contractile cells. Though extremely slow and limited, this is the first appearance of multicellular body contractions without true muscle tissue.

They also evolved silica or calcium carbonate spikes and collagen to retain its shape & rigidity.
They had basic immune systems, ways to recognize what was self and foreign, and could reject cells of another sponge if they entered their system - the biological "sense of self".

They also had sensors that could detect light & dark - sponges contract in bright light, sensors that could detect toxins, changes in chemicals, flow rate and much more - all these inherited from its parent & grandparent species.

They are the first animal, but quite distinct from the rest of the animals to come . One of a kind.

First circulatory fluid - colourless

Page Fungi On Land

Page CombJellies

Another revolutionist, the comb jellies, sponges sibling, who also came from the same mother- the mitochondrial eukaryote. They developed the first nerve nets. The function of thought -sensing, processing, responding, were carried out by the same mechanisms of ion flow and electric impulses as it ever had. But now, with neurons, it became much faster.
The first ones to get true muscle cells, with which they could contract, propel, reshape their body.
The first ones to have mouth and an anus. Mouth came first in their parent species. But until now waste was diffused through the cell walls. But they upgraded to a more efficient one way track. They swallowed prey, digested internally and excreted the waste through the all new anus.
They also had 8 rows of cilial projections that it used to propel and move.
They also were the first ones to start sensing gravity and motion, which brought with it, the first sense of body orientation (front, back, top, bottom) and helped them maintain balance in the 3D environment.
As if all of the above weren't enough to find food, they also became the world's first predators - they started actively executing strategies to actively find and trap prey with sticky cells ejected from its tentacles.Many of them went one more step further & evolved bioluminescence, to startle predators, distract prey and communicate their positions.
They were unique for another reason too - just like all their ancestors and siblings they were also hermaphrodites. But, no one individual produced both sperm and eggs at the same time. And self fertilized if mates were not nearby. Comb jellies - produced sperms and eggs almost everyday and fertilized outside in the water. This helped with population increase when resources were abundant

Circulatory fluid - colourless

Why neurons are not all that its hyped up to be

Page Bilateria

Symmetry was not a feature before now. Organisms were more like blobs without defined top, bottom or left and right. That changed with bilateria. This meant a lot of things - one, they now had defined movement - head first - now that they had a head; They also developed a primitive brain that collected sensory input and distributed signals downwards to the rest of the body through the Central Nervous System.These bilaterians,ancestors to humans and millions of species, inherited the primitive digestive system from their parent species - the mitochondrial eukaryotes. They developed the first circulatory system, with blood & vessels, to carry nutrients and proteins across the body, which was done through diffusion through internal fluid till now. The bilaterians also developed the first hearts to pump blood & allow more control and

efficiency. These little fellows also adopted a wide range of reproductive strategies - external fertilization where sperm and eggs are ejected into water & fertilization happens outside, some adopted internal fertilization. Some were mostly hermaphrodites. Some of them opted for internal copulation, so as to not depend on availability of a male. Some of them developed a master sex determining gene, which set course for its evolution on a specific gender path. The default was female & if this gene were on, then the organism would grow into a male.
Evolving circulatory fluids.

Page The Gender Split **The Birth of Gender**

Specialised gender emerged with Bilaterians.

Why?

Because **eggs** were heavy.
Not just in mass, but also in **meaning**.
They carried nutrients, memory, instructions—
life packed into a single, precious sphere.
And to **build** these, a being needed more:
Internal sensors.
Better signal networks.
Storage. Protection. Strength.

Egg-making wasn't simple.
It asked for **intelligence**—
but the kind that senses, feels, decides, creates.

Thus, emerged the **first females**—
not by decree,
but by the sheer **weight of responsibility**.

And so, nature made a deal.

Let one half carry the load,
and let the other run light.

Sperm was born—cheap, fast, many.
The male strategy was **movement**,
not meaning.
To flood the world with options,
and let selection decide.

Where the **female** became an engine of quality,
the **male** became a storm of quantity.
Where one evolved **complexity**,
the other evolved **speed**.

Both noble.
Both needed.
But not equal in effort.

And once this split happened,
the story of Life was never the same again.

It had found a new rhythm:
the dance of the two—
each carving their path,
each shaping evolution.

Land Green Algae

Cnidarians
From the mitochondrial eukaryotic parents came a species - the cnidarians - sea anemones, jellyfish and corals - Sibling to fungi,sponges and Comb Jellies and the bilateria. They were revolutionary, alright and how! They had a choice in body form. Each cnidarian could assume both of the 2 body forms at different times. One mobile, reproducing sexually through external fertilization, looking much like the older sibling, comb jellies. The other body form- sessile, reproducing asexually forming nodes from themselves, colonial, a higher level multicellular organism that formed colonies of polyps that act like a single organism (sharing nutrients and nerves), much like their other elder sibling, the fungi.
The corals later also innovated secretion of calcium carbonate to form an external skeleton, making the reefs that sustain billions aquatic lives.
But there weren't all benevolent. The cnidarians are also cunning predators. They brought forth a major innovation in animal weaponry – explosive stinging organelles unique to cnidarians. This provided an effective means of prey capture and defense that was unprecedented. Cnidarians became the first animals to wield biochemical toxins for hunting, a strategy that many later animals would also adopt or evolve in parallel.

Colourless gastrovascular fluids - blood.

Recap till now

Protostomes - Mollusks, Annelids, Arthropods

The bilaterians split into the first group, the group for which the mouth formed first in an embryo and then the anus.This dictated where the nervous system was going to be & mouth first meant ,the nervous system on the belly side. They also emerged a new method of cell division when growing into the organism from the embryo stage, where each cell had an allocated location and purpose. This was different from the method so far, where any cell could settle anywhere in the body. The new way was faster. But it lost the ability to regenerate & self heal in gaining speed of development. Each group within the protostome developed different types of limbs, motion and precise muscular control. They could dig, swim and crawl.They also evolved physical armors in the form of shells and tough exoskeletons. Sexual reproduction - internal & external fertilization, many hermaphrodites. Faster life cycles, many offsprings.
Within the protostomes came 3 groups - Mollusks, Annelids(worms) & Arthropods - with similar fundamental structures & features with some variations. Complete digestive system.

Proto blood - circulatory fluids of different colours - beginning of pigment evolution - Red, green, pink -(in annelids), violet & blue (in arthropods and mollusks), - blood flowed across the cells and organs freely. Eventually developed gills and lungs.
Grazing, pollinating, decomposing.

Deuterostomes - Ambulacraria, Chordates - Lancelets

The second group from the parent protostomes were the Deuterostomes - the group for which anus formed first and this meant the nervous system running through the back and not the belly side. The embryo cell division has flexible cell fate, like parents & unlike the protostome sibling. Longer life cycles and fewer offspring.

Two groups formed within the deuterostomes - the ambulacraria & the chordates. Blood colour - colourless, yellow, green, purple, red - blood vessels were developed. Heart became more complex.
They had gills, lungs. Complete digestive system. Mostly sexual reproduction & internal fertilization - gonochorism.
Ambulacarians developed the first endoskeleton, which was refined to notochords by the chordates. They Gave up, head structure - proto brain , eye complex; lost rapid movement for a slow sedentary lifestyle . But these returned in chordates.
Large bodied predators & filter feeders,

Plants Colonize Land

Parasites

Around 512 million years ago, in the crowded Cambrian seas, some arthropods began evolving parasitic lifestyles. This moment marks a radical innovation—not in structure, but in strategy. Cambrian explosion had crowded seas with many large, nutrient-rich filter feeders (e.g., brachiopods, early arthropods). Competition for food was fierce. Some organisms found it easier to tap into others' success than compete directly.

Parasites flipped the script: they **latched onto or invaded** other organisms to **siphon resources**, rather than find or fight for them.
They evolved hooks and adhesive tubes to attach themselves to the host, modified their mouth part to pierce and suck from the host. Simplified body parts and gave up complex sensory organs. They had high rates of reproduction, laying larvae on hosts, since the survival rates were low when born. But of their innovations towards a lazy lifestyle, the most cunning one, was to start generating proteins just like their hosts, inorder to go undetected by the host's immune system.

This wasn't the first time that a species developed parasitic properties too. It had happened many times before, in prokaryotes, protists, fungi, and would happen again, even in plants.

Parasitism is a strategy that has evolved around 200 times across all species, whenever resource was scarce, competition high & had stable host species.

So why are we especially talking about the arthropod parasites? Because these guys are still vampiring on us and have been hugely successful. You might know them too - lice, fleas, ticks, mosquitoes, some flies . While most other parasitic species live within host organisms, these guys live outside the host. They do not only have parasitic property , but carry the genes too , infecting or rather, converting, other species too. Some of these parasites even have developed anesthetics so hosts won't even know when they bite & suck their nutrients. In sheer diversity and ecological range, arthropods out-parasite nearly all other animal groups.
Well, though so much for a lazy lifestyle, they still innovated much.

Cephalopods

Next species emerged from the Mollusks of protostome lineage. The cephalopods were the first mollusks with buoyancy control and they could move vertically in the water. They developed foot, that transformed into tentacles and muscular funnels for jet propulsion. They evolved beaks, arms and tongue with teeth, to win at the predation game.Their parent mollusks had an open circulatory system and simple heart. The cephalopod child evolved a closed circulatory system and multiple hearts.They showed massive expansion in brain and sensory lobes - one of the most advanced among the invertebrates. They evolved the first camera type eye with a lens and focussing abilities, an improvement from the earlier eye spots.

Euthycarcinoids

Almost all of the species that had evolved before them, are still alive to this day. But this group is one of a handful that has now gone extinct. They played a crucial transitional role between fully aquatic marine arthropods and terrestrial arthropods such as myriapods and insects. They appeared around 500 million years ago in the Cambrian and may have been among the first animals to walk on land. Pioneered the legs to enable walking for the first time, an upgrade from the crawling. Pioneered land breathing. They developed thicker exoskeletons and innovated wax coatings to prevent drying on land. Developed new senses with early aerial-tuned eyes (ocelli) and antennae for land sensing. While laying eggs was already an old way, these were the first ones to lay eggs on land.

Too slow in water, too dry on land.
Too fragile for air, too complex for tidepools.
But in that in-between—evolution dreamed.

The 1st Vertebrates

From the chordates of the deuterostomes, came the first vertebrates - the jawless fish - they had a skull and an expanded lobed brain, an upgrade from the simple brains their parent chordates had. Complex eyes. Ears came about for the first time , and helped with balance, just as they do now. Chambered heart, closed circulation. The notochords of Chordates evolved into cartilage based endoskeleton. Larger active swimmers.Gills in chordates were being just used for passive gas exchange. But the jaw less fish gills started actively pumped. Filter feeding was lost for more muscular control. True liver and pancreatic tissues for better nutrients processing. Hemoglobin appeared.Eggs and sperm released into the water

Bony Fish

Complete digestive tract with stomach, intestine, liver, pancreas as in basal jawed fish. Moveable jaws with calcified teeth for grabbing and chewing prey (first time jaws used for true biting). Many bony fish evolved a protrusible jaw for suction feeding (allowing them to slurp prey). Also, pyloric caeca (blind sacs) off the intestine increase nutrient absorption. Acute vision.Precise motor control of fins. Some lineages even developed internal fertilization and live birth (e.g., guppies), showing early viviparity. Kidneys and gills jointly manage waste (ammonia diffuses out at gills). (I) Advanced osmoregulation: marine bony fish drink seawater and excrete excess salt through gills; freshwater bony fish produce very dilute urine to expel water. Nitrogen waste is mostly ammonia (highly toxic but easily flushed in water) – an efficient strategy for aquatic life, with urea production in a few cases when needed (e.g., lungfish during drought).

Scorpion